

A universal action floor needs a fixed action unit and no admitted similarity

A positive action floor shared by all admitted masses and preparations is a dimensionless multiple of a product of the model's fixed constants with action units. A model whose fixed constants admit no such product has floor zero or infinity, and a model whose admitted class is preserved by a one-parameter similarity that rescales the observable has floor zero even when such a product exists. These two elementary criteria decide the recorded action-selection counterexamples from C002 to Q13 before calculation, and they identify the single mass-independent action unit available to classical electrodynamics: $k_e/c = e^2/(4\pi\epsilon_0 c)$, which equals $\alpha\hbar$ and is exactly the excluded angular-action infimum of the relativistic Kepler threshold C052. The selection question therefore separates into the premise fixing a charge unit and the factor $1/\alpha$ between the electromagnetic and quantum action units. This is Q14's consolidation; it promotes no ledger claim.

1. Setting

A model class is specified by fixed constants $c_1, \dots, c_r$, a set of admitted masses, a class of admitted preparations, and dynamics. Each fixed constant has a dimension exponent vector $d_i \in \mathbb{R}^3$ with respect to mass, length and time. Charge enters only through Coulomb energies $q_i q_j / (4\pi\epsilon_0 r)$, so $k = q_i q_j / (4\pi\epsilon_0)$ has dimension energy times length, $d_k = (1, 3, -2)$, and no separate charge dimension is needed. The action dimension vector is $d_A = (1, 2, -1)$. An action observable A assigns a value in $[0, \infty]$ to each admitted solution with its preparation. The attained set $\mathcal{A}(c) \subset [0, \infty]$ collects its values over all admitted masses, preparations and solutions at constants $c = (c_1, \dots, c_r)$, and the floor is

$$g(c) = \inf\{A \in \mathcal{A}(c) : A > 0\},$$

with $g = \infty$ when no positive value is attained.

Two premises are used. **Dimensional homogeneity:** the admitted class, dynamics and observable are defined by dimensionally homogeneous relations. Consequently a change of units $\lambda = (\lambda_M, \lambda_L, \lambda_T) \in (0, \infty)^3$, written $\lambda^d = \lambda_M^{d_M} \lambda_L^{d_L} \lambda_T^{d_T}$, carries the model at constants c to the model at constants $\lambda^{d_i} c_i$ and multiplies every attained action by λ^{d_A} :

$$\mathcal{A}(\lambda^{d_1} c_1, \dots, \lambda^{d_r} c_r) = \lambda^{d_A} \mathcal{A}(c). \quad (1)$$

Universality: the floor depends on the fixed constants only. Admitted masses and preparation parameters are ranged over inside the infimum, so they are absent from the argument list of g . This is obligation 4 of the action target written as a hypothesis. A fixed bath mass, cutoff, apparatus length or supplied coefficient is a fixed constant when the model holds it fixed for all admitted preparations.

2. Theorem A: a universal floor is a multiple of an action product

Theorem A. Under dimensional homogeneity and universality:

- (i) If d_A lies outside the linear span of $d_1, \dots, d_r$, then $g(c) \in \{0, \infty\}$ for every c .
- (ii) If $d_A = \sum_i a_i d_i$, then $\Pi(c) = \prod_i c_i^{a_i}$ has action units and $g(c) = \Pi(c) F(\pi_1, \dots, \pi_s)$, where $\pi_1, \dots, \pi_s$ is a basis of the dimensionless products of the constants and F takes values in $[0, \infty]$. When the d_i are linearly independent there is no dimensionless product and $g = C \Pi$ for one number $C \in [0, \infty]$.

Proof. Equation (1) and universality give $g(\lambda^{d_1}c_1, \dots, \lambda^{d_r}c_r) = \lambda^{d_A}g(c)$ for all λ . Write $\lambda = e^\ell$ componentwise with $\ell \in \mathbb{R}^3$, so $\lambda^d = e^{\ell \cdot d}$. For (i), the orthogonal complement of $\text{span}\{d_i\}$ is not contained in $d_A^\perp$, because in finite dimension $V^\perp \subset w^\perp$ is equivalent to $w \in V$. Choose ℓ with $\ell \cdot d_i = 0$ for all i and $s = \ell \cdot d_A \neq 0$. Then every constant is unchanged and $g(c) = e^{ts}g(c)$ for all real t , which forces $g(c) \in \{0, \infty\}$. For (ii), Π has dimension vector $\sum_i a_i d_i = d_A$, so g/Π is invariant under all unit changes; an invariant function of the constants is a function of a basis of dimensionless products, which is the Buckingham reduction. If the d_i are independent the only dimensionless product is the constant 1. $\square$

The theorem gives the form of a floor and its possible mass dependence, and leaves the positivity of F to dynamics. Its content for selection is the contrapositive of (i): a model that holds no action-dimensional combination of its fixed constants cannot exclude zero universally, whatever its dynamics.

3. Theorem B: an admitted similarity forces floor zero

Theorem B. Fix the constants. Let $\mathcal{S}$ be the admitted set of solutions with their preparations and let $T_a : \mathcal{S} \rightarrow \mathcal{S}$, $a \in (0, \infty)$, satisfy $A(T_a s) = a^\delta A(s)$ for some $\delta \neq 0$. If some $s \in \mathcal{S}$ has $0 < A(s) < \infty$, then $g = 0$, and the attained positive values are also unbounded above. If A takes one common value g on $\mathcal{S}$, then $g \in \{0, \infty\}$.

Proof. $A(T_a s) = a^\delta A(s)$ runs through $(0, \infty)$ as a does. $\square$

The map T_a must preserve the admitted class at fixed constants. The three standard instances are amplitude scaling of a linear dynamics with a preparation class closed under multiplication by a (quadratic actions, $\delta = 2$), velocity scaling of a free or collisional preparation ($\delta = 2$ for a plateau proportional to u^2), and the space–time contraction $q_a(t) = aq(t/a)$ of C056, which is a similarity between different external potentials ($\delta = 1$); C057 is Theorem B for that map with the class enlarged to be closed under it. A dynamics that is linear near an equilibrium supplies T_a asymptotically: the small-radius circles of C058 have $\ell_R \sim \sqrt{mk_s}R^2$ because the confining potential is harmonic at its minimum, and the harmonic oscillator has the exact amplitude similarity.

Theorems A and B are independent necessary conditions. Q02’s regulated oscillator holds q , m , ϵ_0 and c fixed, so $q^2/(\epsilon_0 c)$ is an available action product; its linear response with a preparation class containing every field amplitude admits $E \mapsto aE$, $x \mapsto ax$, $J_* \mapsto a^2 J_*$, and Theorem B gives floor zero. Fixing the amplitude K instead removes the similarity by adding an action-dimensional constant, which is what “supplied coefficient” means throughout the ledger.

4. The recorded countertests and bounds under the two criteria

Result	Fixed constants	Action product	Admitted similarity	Criterion
C002 constant-force variations	m, F, T : (1, 0, 0), (1, 1, −2), (0, 0, 1)	$F^2 T^3 / m$	$\eta \mapsto a\eta$, $\Delta S \mapsto a^2 \Delta S$	B
C024 collision plateau	m, M, ν	none without a fixed speed	incoming velocities $\mapsto \epsilon u$, $H_* \mapsto \epsilon^2 H_*$	A and B
C036 composition	reference m_0, u_0 , λ_0	$m_0 u_0^2 / \lambda_0$	none once the reference is fixed	A: $g = C \Pi$
C056–C057 contractions	m, u , no length	none	$q_a(t) = aq(t/a)$, $\delta = 1$	A and B

Result	Fixed constants	Action product	Admitted similarity	Criterion
C058 fixed force-bounded potential	m, k_s, b, c	$\sqrt{mk_s}b^2$	harmonic core, $\ell_R \sim \sqrt{mk_s}R^2$	B, asymptotic
C060–C061 speed floor, force ceiling	$P_*, v_*, F_{\max}$: (1, 1, −1), (0, 1, −1), (1, 1, −2)	$P_*^2 v_*/F_{\max}$, unique	none: the speed floor breaks amplitude scaling	A: $g = C \Pi$, $C > 0$
C052 relativistic Kepler	k, c : (1, 3, −2), (0, 1, −1)	k/c , unique, mass-independent	none: $r \mapsto \lambda r$, $p \mapsto p/\lambda$, $m \mapsto m/\lambda$, $t \mapsto \lambda t$ fixes L	A: $g = k/c$, $C = 1$
C053 softened core	k, c, a, m	k/c	none	A: $g = (k/c)F(amc^2/k)$, $F \equiv 0$
Q02 radiation balance	$q, m, \epsilon_0, c, w, \Lambda$	$q^2/(\epsilon_0 c), mc^2/w$	field amplitude, $\delta = 2$	B
Q04 topological sector	ρ (energy), c : (1, 2, −2), (0, 1, −1)	none	radius R of the degree-one family	A
Q08 string interference	impedance Z , lengths, c	$Z\ell^2$	pulse amplitude, $\delta = 2$	B
Q13 CHSH witness	probabilities only	none	none needed	A

Each row uses the constants the cited note holds fixed; a row that lists a supplied coefficient among the constants moves to the C036 pattern. The remaining Q-series tests end in a free common multiplier (Q05–Q07) or a dimensionless statistic (Q09–Q12) and fall under the same two criteria. The G-track results are in the same position on the other side: a quantum Hamiltonian with couplings in energy units and a lattice spacing has no action product without a fixed time or a supplied $\hbar$, which is the recorded status of C126 and G05.

5. Which classical constants supply a mass-independent action unit

Proposition. Let the fixed constants be $k_e = e^2/(4\pi\epsilon_0)$, c and G , with dimension vectors (1, 3, −2), (0, 1, −1) and (−1, 3, −2). The unique product with action units is k_e/c . Without k_e there is none; admitting a mass m adds Gm^2/c and its mass-dependent relatives.

Proof. The three vectors are linearly independent, so the exponents (a, b, γ) with $a(1, 3, −2) + b(0, 1, −1) + \gamma(-1, 3, −2) = (1, 2, −1)$ are unique. Adding the length and time components gives $a + \gamma = 1$; the mass component gives $a - \gamma = 1$; hence $a = 1$, $\gamma = 0$, $b = -1$. With G and c alone the mass component forces $\gamma = -1$, and then the length and time components are inconsistent. With m admitted, Gm^2/c has dimension $(-1 + 2, 3 - 1, -2 + 1) = (1, 2, -1)$. $\square$

In quantum units $k_e/c = \alpha\hbar$ with $\alpha = e^2/(4\pi\epsilon_0\hbar c) \approx 1/137$. Thus the fixed constants of classical mechanics, electrodynamics and gravitation admit exactly one mass-independent action, and it exists because the charge is a fixed unit. Charge quantization is the classical premise that supplies an action unit; the value it supplies is α times the quantum one. A fixed length together with a mass unit and c would supply another, $m_0 c \ell_0$, at the price of a mass unit and a fundamental length, both absent from the models in use.

6. The relativistic Kepler threshold through the five obligations

C052 already realizes the unique electromagnetic unit. For one body in the singular potential $-k/r$ with $k = k_e$ for two elementary charges, regular bound orbits exist exactly for $|L| > k/c$.

1. **Definition and units.** The angular action $|L|$ of regular bound orbits; k/c has action units.
2. **Exclusion of zero.** The excluded infimum k/c follows from the singular core, finite speed and binding (C052); a softened core removes it (C053). The premise excluding zero is the $1/r$ singularity at fixed k and c .
3. **Convergence.** No limit is required: the threshold is exact at fixed constants, and closes only as $c \rightarrow \infty$ or $k \downarrow 0$.
4. **Universality.** The threshold is independent of the mass, as the explicit formula and the similarity in Section 4 show, and it bounds every regular bound preparation. It scales with the charge product: for a centre of charge Ze it is $Z\alpha\hbar$. The model excludes recoil, radiation and a dynamical second body.
5. **Quantum role.** None is derived. The floor is $\alpha\hbar$, so a factor $1/\alpha$ separates it from the quantum unit, and no premise of the model supplies that factor.

Remark, audited. The quantum Coulomb problems place their collapse thresholds at $Z\alpha = l + 1/2$ (Klein–Gordon) and $Z\alpha = |\kappa| = j + 1/2$ (Dirac). Both are the classical condition $|L| > k/c$ with $|L|$ measured as $\hbar(l + 1/2)$ or $\hbar|\kappa|$, and C052’s circular energies at $|L| = \hbar|\kappa|$ are exactly the Dirac levels with zero radial quantum number. The Q14 review derives this from C052’s energy identity, whose $1/r^2$ coefficient $c^2\ell^2 - k^2$ is what the wave equations’ indicial equations test, and records the primary sources; it is the Sommerfeld–Dirac coincidence of 1916 and 1928. The only quantum input is the unit of angular action at the plunge boundary, so the factor $1/\alpha$ of item 5 says that the Dirac ground state reaches that boundary at $Z = 1/\alpha$.

7. Strategic consequence

Before calculating a selection candidate, list its fixed constants and check the two criteria. If no action product exists, or an admitted similarity rescales the observable, the candidate can supply at most a spectral shape (Q02), a conditional bound (C060) or a dimensionless witness (Q13), and the calculation may be skipped or bounded accordingly.

The one classical route with a mass-independent positive floor is electromagnetic: a fixed charge unit, relativity and a singular Coulomb core give $\alpha\hbar$. The open premises of the selection track are therefore the origin of a fixed charge unit and the factor $1/\alpha$; both lie outside classical dynamics, and the second is the measured fine-structure constant. A return to a physical selection mechanism should start from a model whose constants are e , ϵ_0 , c and masses, with a nonlinear coupling that breaks amplitude similarity, for example the radiating two-body Coulomb problem that Q02’s review names as its remaining dependency; its radiation-reaction length $k_e/(mc^2)$ and action unit k_e/c are then both fixed. Linear reservoirs, contraction-closed classes and dimensionless statistics remain parked by the criteria above.

Proof status: elementary written proofs above; the dimensional reduction is the established Buckingham theorem. Literature status: dimensional analysis and mechanical similarity are established methods, and the identification $k_e/c = \alpha\hbar$ is standard; no novelty is asserted. A bounded librarian comparison and written review are required before any ledger promotion.